

Endometriosis is a chronic disease affecting women of reproductive age. It is characterized by the presence of endometrial-like tissue outside the uterus, which induces a local inflammatory response [1]. Common symptoms include chronic pelvic pain, fatigue, congestive dysmenorrhea, heavy menstrual bleeding and severe dyspareunia. Endometriosis affects 47% of infertile women [1]. It is a public health problem affecting approximately 176 million women worldwide, or 5–10% of women of reproductive age [2]. New evidence suggests that endometriosis is associated with several chronic diseases, including atherosclerosis, coronary heart disease (CHD), dyslipidemia, hypertension, autoimmune disease, and even gynecological cancers [3]. Endometriosis has been

linked to systemic chronic inflammation, heightened oxidative stress, and an atherogenic lipid profile [4, 5]. In the pathogenesis of atherosclerotic CAD, inflammation, oxidative stress and an atherogenic lipid profile play a key role [6–8]. Chronic systemic inflammation contributes to vascular insult and atheromatous plaque progression [9]. Increasing evidence supports the idea that reactive oxygen species contribute to the process of atherogenesis, as important mediators of signaling pathways that lead to vascular inflammation. Therefore, the presence of endometriosis may promote coronary artery atherosclerosis formation and progression, increasing the risk of CHD [8].

Cardiovascular disease (CVD) is a general term for conditions that affect the heart and blood vessels. CVD includes various disorders such as coronary artery disease (CAD), hypertension, high cholesterol, atherosclerosis, and diabetes. Also, most biomarkers in the pathobiology of endometriosis show a convincing association with CVD [10]. CVD is one of the main causes of death in women, according to 1 death out of every 3 deaths per year [11]. Meanwhile, endometriosis affects 1 out of every 10 women in fertility [10]. Statistics show

that the relative risk (RR) in endometriosis patients for developing hypertension is 14% increased, hypercholesterolemia is 25% increased risk [12] and ischemic heart disease (IHD) is 40% increased, and myocardial infarction (MI) is 52% increased risk [13]. The association between endometriosis and CVD has been investigated in cohort and case-control studies. However, the findings have been inconsistent, with some studies suggesting a positive association between endometriosis and CVD complications, while others have found no significant correlation. This lack of consensus has resulted in uncertainty among healthcare professionals regarding the optimal management of patients with endometriosis. Moreover, there have been no systematic reviews or meta-analyses conducted thus far to comprehensively evaluate this association. To address these knowledge gaps and establish a more reliable understanding of the link between endometriosis and CVD, a systematic review and meta-analysis has been undertaken.

Methods

The present study was a systematic review and meta-analysis performed based on the six basic steps of search syntax and search strategy, screening, selection, data extraction, quality assessment, and meta-analysis according to the structure of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [14].

The protocol for this review was registered in PROSPERO (CRD42023398887).

Eligibility criteria

The eligibility criteria were defined using the PECOT framework: Population (women), Exposure (endometriosis), Comparison (individuals without endometriosis), Outcome (incidence of cardiovascular diseases or hypertension), and Time (studies published from January 2000 to April 2023). Studies eligible for inclusion were cohort or case–control designs reporting relative risks (RRs) or equivalent effect measures with 95% confidence intervals (CIs). Exclusion criteria included case reports, case series, clinical trials, letters to editors, nomograms, comparative studies, review articles, cross-sectional studies, and studies without full-text availability in English. Articles were selected systematically through the PECOT structure to ensure relevance and alignment with the study objectives.

Information sources and search strategy

A comprehensive literature search was conducted in PubMed (Medline), Embase, Scopus, and Web of Science (ISI) to identify relevant studies published from January 2000 to April 2023. The search strategy was designed using Medical Subject Headings (MeSH) terms and synonyms for "endometriosis" (e.g., "endometrioma") and "cardiovascular diseases" (e.g., "adverse cardiac event", "major adverse cardiac events", "cardiovascular abnormality"). The full search syntax for each database is available in Supplementary Material.

Selection process

Search results were imported into Endnote8 software, and duplicates were removed both automatically and

manually. Two independent reviewers (SP and RN) screened the titles and abstracts for eligibility, with any discrepancies resolved by a third reviewer (YM). Full text articles were subsequently assessed for inclusion. Reasons for exclusion during full-text screening were documented.

Data collection process

A structured data extraction form, developed by the research team, was used to retrieve information from eligible studies. Extracted data included author names, publication year, study design, population characteristics, sample size, exposure definitions, effect estimates (RRs), and their 95% CIs. Data were extracted independently by two reviewers, with discrepancies resolved through discussion with a third reviewer.

Data extraction

After selecting the articles in the screening stage based on their titles, abstracts, and full texts according to the inclusion and exclusion criteria, the checklist prepared with the opinions of the experts was used to retrieve the data of the selected articles based on the research objective. The checklist dealt with the name of the author(s), type of study, year of publication, total sample size, and effect size with a 95% confidence interval (95% CI). Two authors independently extracted all data based on the desired form, and in case of any discrepancies, the results were reviewed by the third author.

Quality assessment

The quality of the studies was evaluated using the New

castle–Ottawa Scale (NOS) [15]. Based on this scale, a maximum of 9 points were awarded to each study, which included 4 points for the selection of participants, 2 points for comparability, and 3 points for outcome ascertainment. Two authors have independently done quality assessments based on the desired checklist, and in case of any discrepancies, the results were reviewed by the third author.

Statistical analysis

All included studies in this meta-analysis were cohort studies that reported RRs and 95% CIs. A pooled analysis of the logarithm of the RR and its 95% CI was conducted. The fixed-effects model was applied for analyses with low heterogeneity ($I^2 < 50\%$), while the random-effects model was used when high heterogeneity was observed

($I^2 \geq 50\%$). Cochran's Q test and the I^2 statistic were employed to evaluate heterogeneity and variance among the included studies. Subgroup analyses, sensitivity analyses, and funnel plot assessments were not performed due to the limited number of included studies, which restricted the ability to explore potential sources of heterogeneity or assess publication bias effectively. All statistical analyses were conducted using STATA 17 software.

Results

Our initial search yielded 1849 articles from all databases. Among these, 755 studies were excluded due to duplication, and 1094 remained. After the screening that was done in terms of title and abstract, only 15 studies

remained. Among the reasons for excluding the studies were the irrelevant title of the study results as well as the method and effect sizes. Also, we only selected studies that were case-control or cohort (Fig. 1). In the final screening based on full text, only six studies were selected for statistical analysis, all of which were cohorts. It should be noted that the statistics obtained from each of the selected studies were related to CVD or hypertension, so we categorized the results into CVD and hypertension to solve the challenge (Table 1).

CVD risk in women with endometriosis

Out of the 6 selected studies, 5 studies were analyzed in the CVD category, whose details and characteristics are presented in Table 1. The statistics obtained from each of the studies were as follows: Chun-hui Wei et al. [16] (RR: 1.52; 95% CI: 1.20—1.84), Powell. M. J et al. [17] (RR: 0.48; 95% CI: 0.18—0.77), K Okoth et al. [13] (RR: 1.24; 95% CI: 1.12—1.36), Hsin-Ju Chiang et al. [18] (RR: 1.19; 95% CI: 1.00—1.39), Pei-Chen Li et al. [19] (RR: 1.34; 95% CI: 1.22—1.47). Finally, by combining results, the pooled RR was calculated as 1.23 (RR = 1.23; 95% CI: 1.16—1.31;

I² = 87.15%). This indicates a 23% increase in the risk of cardiovascular disease in women with endometriosis compared to those without endometriosis (Fig. 2).

Hypertension risk in women with endometriosis

Out of 6 selected studies, 2 studies had reported a risk of hypertension in women with endometriosis. K Okoth et al. [13] reported a relative risk of 1.12 (RR = 1.12; 95%

CI: 1.07—1.17) for hypertension in women with endometriosis. According to a study conducted by Fan Mu et al. [12], women with endometriosis had a relative risk of 1.14 for hypertension (RR = 1.14; 95% CI: 1.10—1.18). A meta-analysis of these two studies yielded a pooled RR of 1.13 (RR = 1.13; 95% CI: 1.10—1.16; I^2 = 0.00%), indicating that women with endometriosis have a 13% higher risk of hypertension compared to those without the condition (Fig. 3).

Discussion

The primary aim of this meta-analysis was to investigate the association between endometriosis in women and their risk of developing CVD, and our secondary aim was to investigate the association between endometriosis in women and their risk of developing hypertension. The results of this meta-analysis indicate that women with endometriosis may have a higher risk of developing CVD and hypertension compared to women without endometriosis. However, these findings should be interpreted cautiously, as further research is needed to confirm the association and understand the underlying mechanisms. To our knowledge, this is the first meta-analysis worldwide to look at the association between endometriosis and CVD in women. Previous studies published to date have reported controversial results regarding this association. These findings are consistent with previous studies, although the exact nature of this correlation is still uncertain. The proposed biological mechanisms, such as chronic inflammation, oxidative stress, and the impact

of estrogen, offer a logical explanation. However, further research is needed to confirm these pathways and establish

a cause-and-effect relationship. Moreover, shared lifestyle risk factors may contribute to this observed association, adding complexity to the link between endometriosis and cardiovascular outcomes. Given these complexities, although our findings are noteworthy, it is important to interpret them with caution. Future research should focus on clarifying these associations and investigating potential interventions. Endometriosis is a condition in which tissue similar to the lining of the uterus grows outside the uterus. The most common sites for this condition are the pelvic area and the ovaries, and it is more common in women under the age of 40. Based on previous studies, it is estimated that 11% of women in the United States, or about 6 million women, have this condition, and the most common age range for developing it is between 30 and 40 years old. Chronic systemic inflammation, increased oxidative stress and atherogenic lipid profiles associated with endometriosis may biologically link this condition to cardiovascular disease. Another clinical rationale for this association may be genetic factors [20]. In previous European studies, several independent loci, such as ANRIL on chromosome P219, have shown a significant association with endometriosis [21, 22]. In addition, studies have shown a correlation between the occurrence of CVD and this genetic locus, which may support a link between endometriosis and CVD. In addition, previous studies have shown that long non-coding RNAs, such as Chr9p21 RNA, play a role in cell proliferation, adhesion, and apoptosis and ultimately increase the risk of developing endometriosis [23]. In addition to this clinical justification, the association between endometriosis and the incidence of CVD can also be considered from another perspective. Endometriosis is generally a women's disease in which risk factors such as physical inactivity, unhealthy diet, alcohol consumption and smoking play an important role in its development [24, 25]. These factors are also considered to be important contributors to the occurrence of CVD and related outcomes, and the presence of common risk factors between these two conditions may also contribute significantly to their relationship [26]. Endometriosis is a disorder in which the tissue that borders the uterus (endometrium) develops outside the uterus, most commonly in the pelvic region. Estrogen is known to increase endometrial tissue growth and is implicated in the menstrual cycle [27]. Estrogen has immunomodulatory properties and may alter inflammatory processes in the body. Some speculate that estrogen may have a role in increasing inflammation within endometriotic lesions, which may contribute to the development or progression of CVD [28]. Furthermore, estrogen has been proven to have both beneficial and harmful effects on blood arteries. Speculation may center on how estrogen levels, particularly in women with endometriosis, alter endothelial function, arterial stiffness, or the production of atherosclerotic plaques, all of which are associated with CVD. Estrogen can affect the coagulation system, increasing the likelihood of blood clot formation. Consider whether estrogen-related changes in coagulation pathways may contribute to an increased risk of cardiovascular events in women with endometriosis [27]. In the line of this meta-analysis results, previous published studies showed that endometriosis and CVD, such as atherosclerosis (hardening and narrowing of the arteries) and hypertension (high blood pressure), have been linked in several studies [29]. It's important to remember that the exact nature of these associations and the underlying mechanisms are still poorly understood [10]. Endometriosis and atherosclerosis are usually thought to be separate entities, with endometriosis primarily affecting young reproductive-age women and atherosclerosis being an aging-related condition [30]. Recent findings have revealed cellular and molecular similarities between the two illnesses [31]. In general, the association between endometriosis and cardiovascular disease has

been relatively challenging and not tested in published studies, as endometriosis frequently affects young women while cardiovascular disease primarily affects older women [32]. However, recent research has shed light on the cellular and molecular interactions between the two disorders. Chronic inflammation, increased oxidative stress, endothelial dysfunction, and increased cellular proliferation are all characteristics shared by cardiovascular disorders and endometriosis [33, 34]. There are opportunities to improve the management of endometriosis while preventing CVD by understanding the common pathways between endometriosis and atherosclerosis [35]. Treatments for endometriosis can be tailored to reduce cardiovascular risk by targeting chronic inflammation, oxidative stress and hormonal pathways. These pathways are similar in both conditions. Furthermore, recognition of endometriosis as a potential risk factor for CVD allows for early cardiovascular risk assessment and intervention, with a focus on lifestyle changes and personalized care to reduce cardiovascular risk in endometriosis patients. This meta-analysis exhibited notable strengths and limitations, shedding light on the association between endometriosis and CVD, as well as hypertension. Notably, it stands as the pioneering meta-analysis conducted on this subject matter. However, it is crucial to acknowledge that the limited number of studies included in this analysis represents a significant constraint. This limitation hindered the possibility of conducting subgroup analyses, which could have explored key variables such as age, duration of endometriosis, the specific methods employed for detecting endometriosis, comorbidities and the types of treatments administered. Another main limitation of this meta-analysis is the lack of detailed reporting on the impact of key confounding variables and comorbidities, such as diabetes, hyperlipidemia, smoking, sex, and body mass index, in the included studies. Although these factors were considered during data extraction, the primary studies did not consistently provide stratified results or adjusted analyses based on these variables. This limitation hinders the ability to fully assess the independent association between endometriosis and CVD. By expanding the scope of the study and incorporating additional research, future meta-analyses can address these limitations and further elucidate the complex relationship between endometriosis, CVD, and hypertension. In future research endeavors, it is imperative to conduct primary studies utilizing cohorts or case-control designs with ample sample sizes. These studies are crucial for gaining a nuanced understanding of the exact association between endometriosis and hypertension, as well as cardiovascular diseases. It is essential that these primary investigations meticulously account for influential variables affecting the relationship between endometriosis and cardiovascular conditions, encompassing factors such as age, underlying medical conditions, and past therapeutic interventions.